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Unit – 5 : Session – 9 : SLO – 2 | Instruction Encoding and Addressing Modes

■ I. Multiple Choice

1. B) Represent instructions in binary.
2. B) Fixed-length encoding — RISC's uniform length simplifies fetch/decode and pipelining.
3. C) How operands are accessed.

■ II. Fill in the Blanks

1. machine code — encoding translates assembly into binary machine code.
2. fixed — RISC uses fixed instruction length.
3. is accessed — the addressing mode specifies how an operand is accessed.

■ III. True or False

1. TRUE — decode complexity from encoding directly affects pipeline efficiency.
2. FALSE — variable-length instructions complicate and slow decoding.
3. TRUE.

■ IV. Match the Following

- A) Opcode — 2) Operation to perform.
B) Operand Specifier — 3) Register/memory location info.
C) Addressing Mode — 1) How the operand is accessed.
D) Instruction Length — 4) Bits used per instruction.

■ V. Numerical Answers

1. 128 — $2^7 = 128$ possible opcodes with a 7-bit opcode field.
2. 26 bits — $32 - 6 = 26$ bits remain for operands/mode fields.
3. 32 — $2^5 = 32$ registers are addressable with 5-bit specifiers.